

Project Initialization and Planning Phase

Date	30 June 2026
Team ID	*****
Project Title	Smart Lender – AI-Powered Loan Eligibility Prediction System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed AI-Powered Loan Eligibility Prediction System aims to simplify and automate the loan approval process using machine learning techniques. The system analyzes applicant information such as income, loan amount, credit history, and employment details to predict loan eligibility accurately. It reduces manual effort, improves decision-making, minimizes processing time..

Project Overview	
Objective	The primary objective of this project is to develop an intelligent loan eligibility prediction system using machine learning. It helps financial institutions make faster, accurate, data-driven loan approval decisions.
Scope	The project comprehensively assesses and enhances the loan approval process, incorporating machine learning for a more robust and efficient system.
Problem Statement	
Description	The traditional loan approval process is time-consuming and depends on manual verification, leading to delays and inconsistent decisions.
Impact	Manual processing increases workload, slows loan approvals, and may reduce the accuracy of decision-making.
Proposed Solution	
Approach	A Random Forest machine learning model integrated with Flask predicts loan eligibility quickly using applicant details..
Key Features	• Fast loan prediction & Reduced manual effort • High prediction accuracy & Real-time decision support • User-friendly web interface.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Processor required for model training & application execution.	Intel Core i5 / AMD Ryzen 5 or higher
Memory	RAM requirement	8 GB
Storage	Storage for source code, datasets, models, and reports	50 GB SSD or above
Software		
Frameworks	Web application frameworks	Flask
Libraries	Machine learning and data processing libraries	scikit-learn, pandas, numpy, matplotlib, Joblib
Development Environment	IDE used for developers	Visual Studio Code, Jupyter Notebook
Data		
Data	Loan applicant dataset used for training and testing.	Kaggle Loan Prediction Dataset (.csv)